

Before Lecture 7

- A perfectly competitive market is an extreme form of market competition; in this market, there are numerous firms, all of which are price takers, and there are no entry barriers. While competition leads to production efficiency, it also eliminates profits.
- A monopoly is another extreme form of market competition; in this market, the entry barriers are extremely high, so there is only one firm, and because there is only one firm, it has the power to influence prices.
- For a firm having the power to influence commodity price, what should its production decision look like?

Lecture 7: Monopoly

7-1: The characteristics and reasons for the existence of a monopoly.

- **The characteristics of a monopoly:**

- (1) This market has only one firm**

- The firm may sell goods, such as water utility firms
 - The firm may provide services, such as the Taiwan Stock Exchange.
 - The number of consumers this firm serves can be large or small.
As long as there is only one firm providing a particular product or service to a group of consumers, these consumers are facing a monopoly.

- (2) The product or service has no close substitutes**

- If a product or service has close substitutes, it means consumers can purchase other products or services in the market to satisfy similar desires. In such cases, it is difficult for the product or service to maintain a monopoly.

- (3) Barriers to market entry exist**

- If a product market does not allow new firms to enter, it indicates the existence of barriers to entry in that market.

- **The reasons for the emergence of a monopoly:**

If a monopoly is not profitable, firms should leave the market in the long run, and monopolist will not exist. If a firm is profitable, the product market should attract other firms to enter and share the profits; in this case, monopolist would also cease to exist.

The reason for the existence of monopolist lies in the fact that the product market is profitable, but new firms are unable to enter!

There are four reasons leading to the existence of monopoly firms:

(1) The firm's production benefits from economies of scale.

(2) There are significant barriers to entry in the product market.

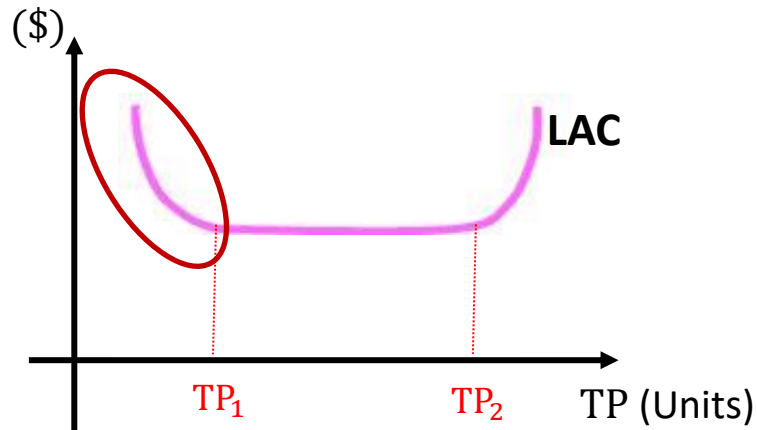
(3) Rare production factors are exclusively owned.

- Rare

(4) The firm possesses unique production techniques .

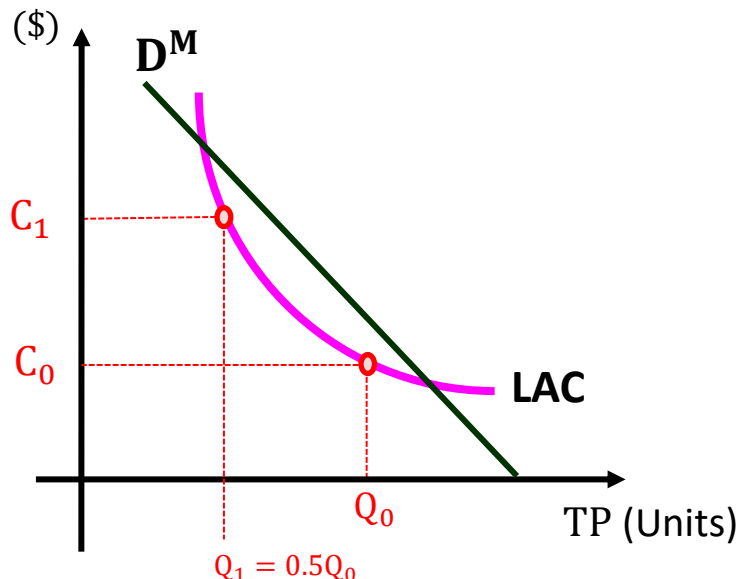
- The first reason for the emergence of a monopoly:

The firm's production benefits from economies of scale



- Recall:** A firm's production benefits from economies of scale, meaning that as the firm's total production (TP) increases, the long-run average cost (LAC) decreases, as shown in the red-circled area in the figure on the left.

- In the bottom-left diagram, the dark green line D^M represents the market demand curve



- When two firms with economies of scale operate in the market, and the product market only contains these two firms, the larger-scale firm ($TP=Q_0$) has a lower average production cost ($TP=Q_1$) compared to the smaller-scale firm ($C_0 > C_1$), even when other conditions are identical. In the long term, the larger-scale firm is more likely to develop strategies to attract customers, making it difficult for the smaller firm to compete, ultimately forcing the smaller firm out of the market.

- As a result, the larger-scale firm naturally forms a monopoly.

- The second reason for the emergence of a monopoly:

Barriers to market entry exist

- Common barriers are often created by the government, with patents and project licenses being the most common types.

(1) Patent

- The government ensures firms' post-production profits through a patent system, encouraging pre-production R&D activities, which facilitates the emergence of new products.
- Patents are time-limited, and firms with patents become legally protected monopolists during this period.

(2) Project License

- The government grants specific firms legal monopoly status through laws or administrative orders.

Example: The Public Monopoly Bureau in earlier years.

- In addition to the legislative creation of market entry barriers, existing firms in an economic society sometimes use various methods to crowd out or block other firms from entering the market. Such actions violate the Fair Trade Act.

7-2: The optimal production strategy for a monopolist under a single pricing strategy

- The firm's objective: Maximizing the firm's profit.
- A monopolist is the sole supplier in a specific product market, so the demand it faces is the total demand from all consumers in the market for that product.
- As long as it is profitable, the firm will continue production (a description relevant to the firm's long-term behavior).

- If the market price of product X is P_X , and since the firm faces the entire market demand for product X, if the market demand curve for this product, D_X^M , is given as: $P_X = 16 - Q_X^{D.M}$

Then the firm's (short-term) total sales volume at the product price P_X is: **$Q_X^{D.M} = \text{STP (Units)}$**

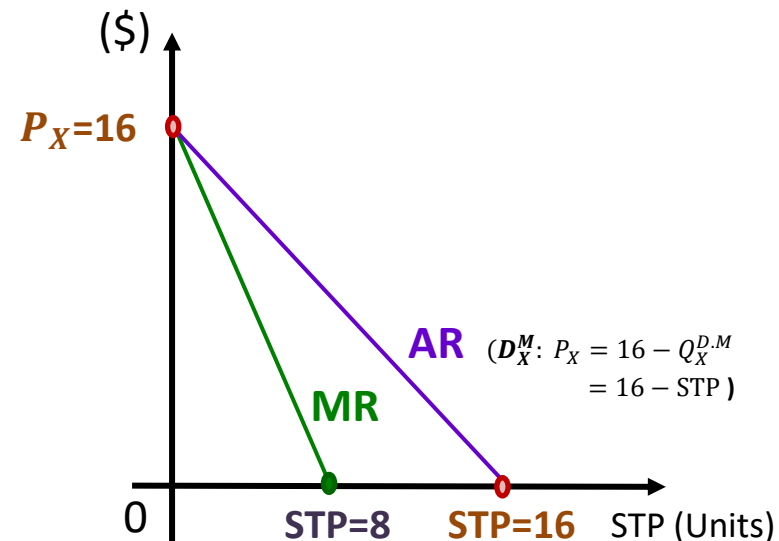
The firm's total revenue (TR) can be expressed as:

$$\begin{aligned} \text{TR} &= P_X \times \text{STP} \\ &= (16 - Q_X^{D.M}) \times \text{STP} \\ &= (16 - \text{STP}) \times \text{STP} \\ &= 16\text{STP} - \text{STP}^2 \end{aligned}$$

The average revenue (AR) can be written as:

$$\text{AR} = \frac{\text{TR}}{\text{STP}} = P_X = 16 - \text{STP}$$

The marginal revenue (MR), $\Delta\text{TR}/\Delta\text{STP}$, can be expressed as: $\text{MR} = 16 - 2\text{STP}$



Perfectly Competitive Market

1. There are many firms in the product market; firms are price takers.

2. Assume $P_X = P_{X.0}$

Let $P_{X.0} = 8$ and $STP_0 = 4$

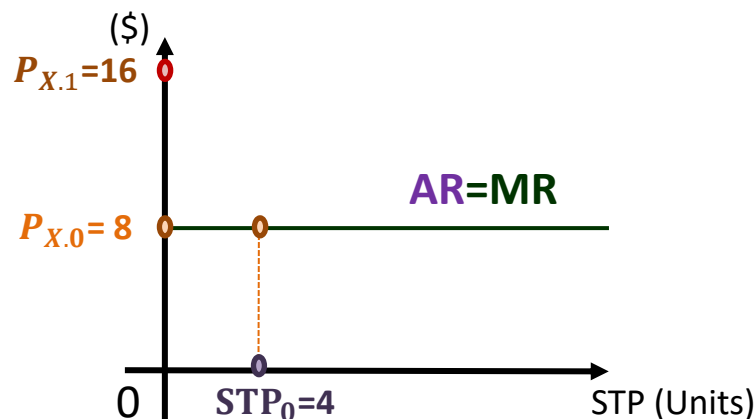
Then

$$TR_0 = P_{X.0} \times STP_0 = 32$$

$$AR_0 = TR_0 / STP_0 = P_{X.0} = 8$$

$$MR_0 = 8$$

3. Horizontal AR line



Monopoly

1. There is only one firm in the product market; the firm faces all the demand in the product market, so the price of the product is determined by the demand curve.

2. Assume $P_X = P_{X.0} = 8$

Then the corresponding quantity demanded is:

$$\underline{Q_{X.0}^{D.M} = 16 - 8 = 8}$$

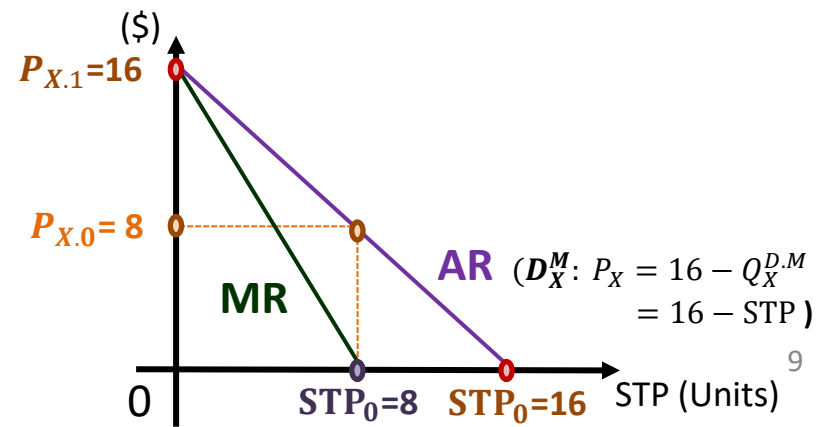
In monopoly, if $Q_X^{D.M} = STP$,
then $Q_{X.0}^{D.M} = STP_0 = 8$

$$TR_0 = P_{X.0} \times STP_0 = 64$$

$$AR_0 = TR_0 / STP_0 = 8$$

$$MR_0 = 16 - 2STP_0 = 0$$

3. AR line is the demand curve



Perfectly Competitive Market

Monopoly

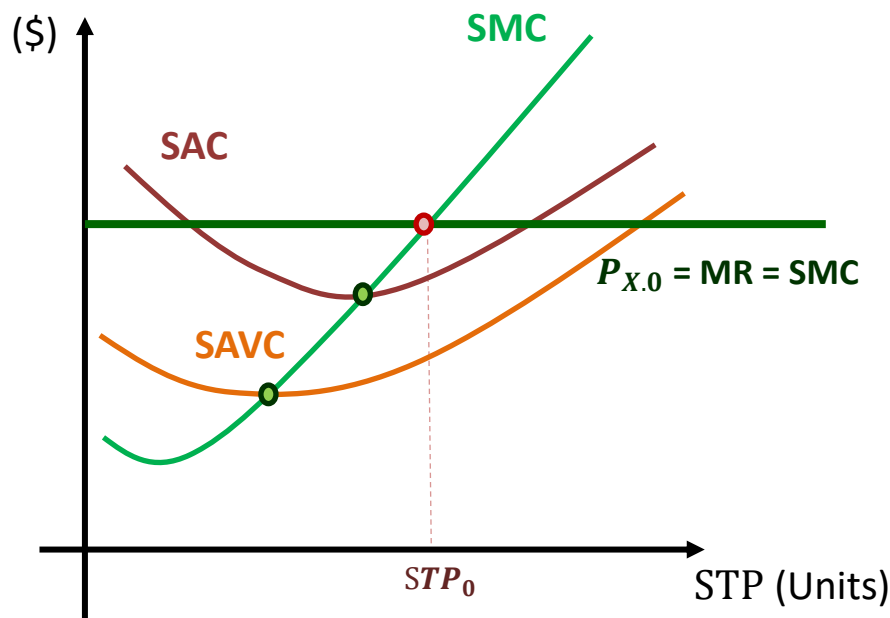
4. Recall:

SMC : the increase in cost for producing one more unit of the product during the short-term period

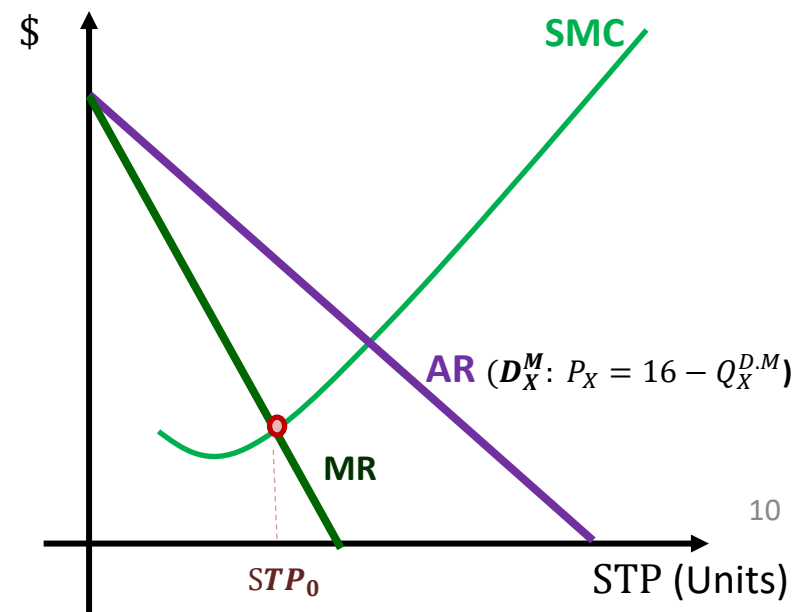
MR: the increase in total revenue for producing one more unit of the product.

A rational firm will stop production when $MR < SMC$, because the marginal profit for producing each additional unit is negative. Conversely, the firm will continue production when $MR > SMC$, because the marginal profit is positive; thus a rational firm should produce until $MR = SMC$, and then stop. This is because the STP at that point maximizes short-term production profit (though the maximum production profit does not guarantee it is greater than 0).

5. A firm will produce $STP = STP_0$
corresponding to $P_{X,0} = MR = SMC$



5. A firm will produce $STP = STP_0$
corresponding to $MR = SMC$:



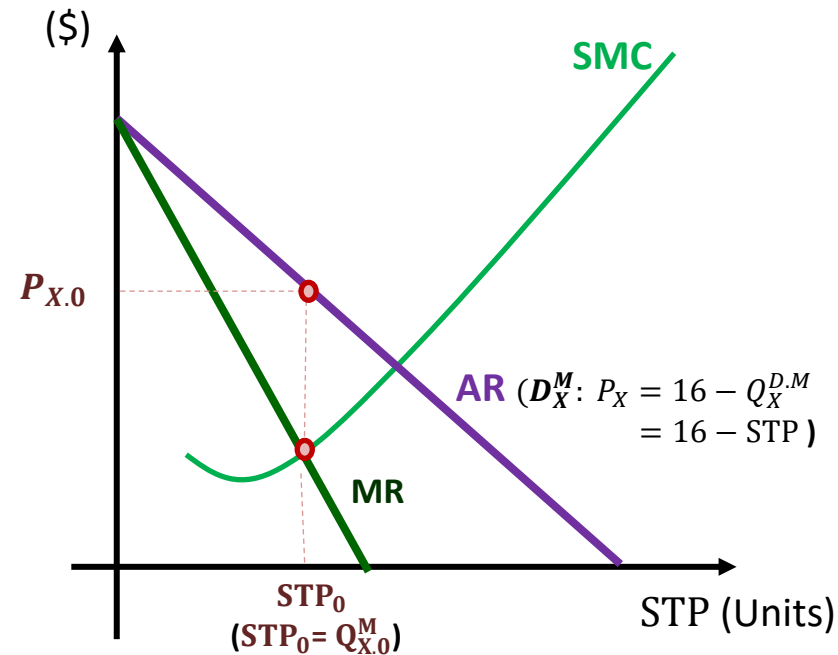
- The optimal output and pricing of a monopolist

- When the total market demand for product X equals the monopolist's STP, its AR will equal the market demand curve D_X^M .

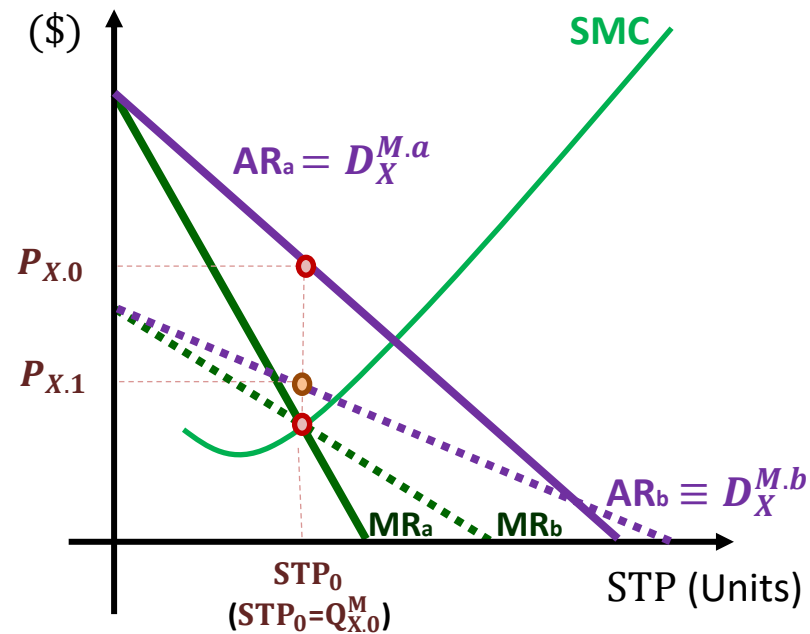
- When the monopolist's $MR = SMC$, the firm's (short-run) production profit reaches its maximum, and the corresponding total output of the firm is STP_0 .

- When the firm's optimal production quantity is STP_0 , and $STP_0 = Q_{X,0}^M$, the corresponding product market price is $P_{X,0}$

- Therefore, the actual approach of a monopolist is: first, set the $P_{X,0}$ for product X, because the monopolist knows that the market will respond with a total demand of $Q_{X,0}^M$ at this price. As long as the monopolist ensures $STP_0 = Q_{X,0}^M$, it can determine an appropriate cost structure (i.e., STC, SAC, and SMC) such that STP_0 satisfies the condition $MR = SMC$, making STP_0 the total output that maximizes production profit."



- Supply Curve (Supply Function) of a Monopolist
 - Function: independent and dependent variables are 1-to-1 relationship
 - As long as $MR > SMC$, the firm is willing to supply products to meet market demand until the total output reaches the state where $MR = SMC$, at the optimal STP.
 - However, the same optimal STP_0 can be derived from the condition $MR_a = SMC$ or from $MR_b = SMC$, each corresponding to different market demand curves $D_X^{M.a}$ and $D_X^{M.b}$.
 - At the same time period, in a monopoly, a firm can have the same optimal total output (sufficient to meet total market demand) corresponding to two different product pricing levels, which are determined by the two demand curves $D_X^{M.a}$ and $D_X^{M.b}$. This results in a contradiction!

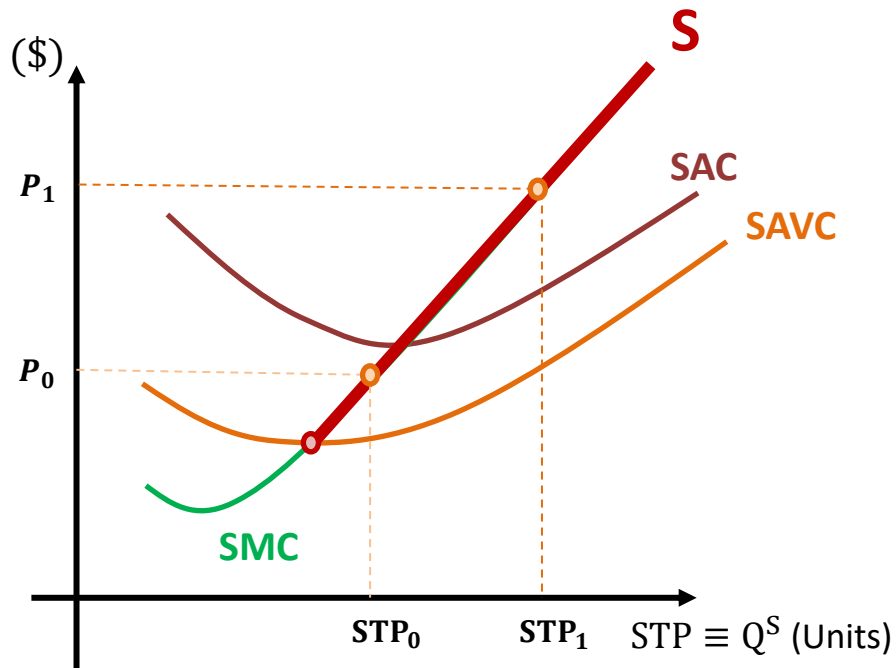


- In a monopoly, the supply curve of a monopolist does not exist, so it is impossible to describe the relation between supply quantity and price!

This is because the definition of quantity supplied is the amount a firm willing and able to sell when facing various prices. However, a monopolist does not need to face prices but can instead determine the price; therefore, the monopoly does not fit the situation described by the supply curve!

Perfectly Competitive Market

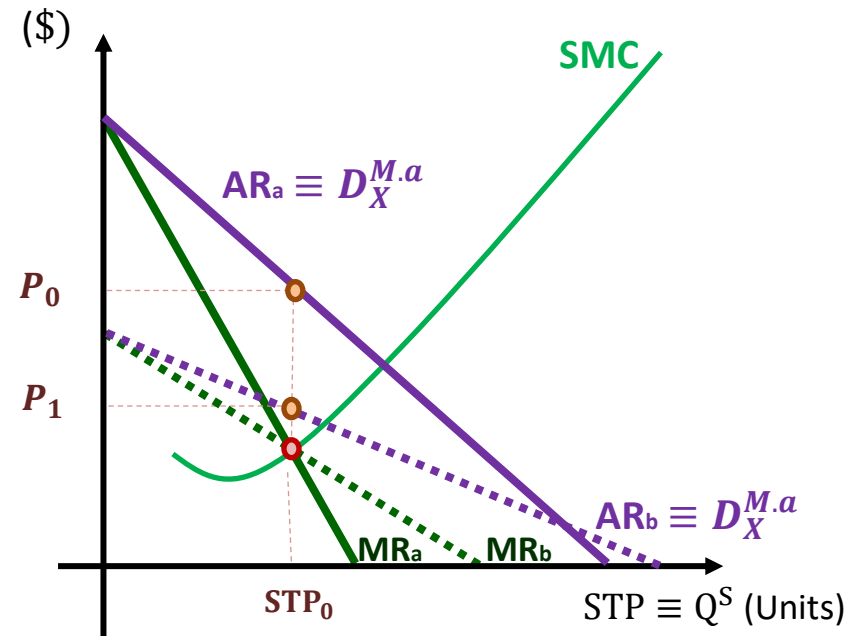
6. In a perfectly competitive market, the short-run supply curve of a firm exists.



The total output (STP) of a firm and the price (P) have a one-to-one relationship.

Monopoly

6. In a monopoly market, the supply curve of a firm does not exist.



The total output (STP) of a firm and the price (P) have a one-to-many relationship, which does not satisfy the definition of a function. 13

• A monopolist's optimal decision in the short run (STP and P)

- In the short run, as long as the firm's short-run total profit (STPf) > 0, the firm will remain in the market.

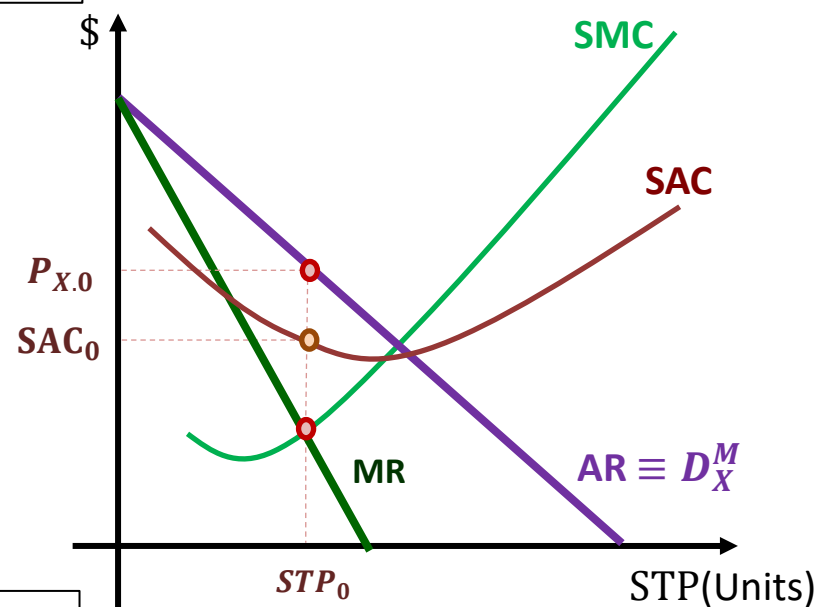
$$\text{STPf} = (P - \text{SAC}) \times \text{STP}$$

- In a monopoly, the firm's optimal output corresponds to $\text{MR} = \text{SMC}$, which determines the STP; the price corresponding to this STP on the AR curve is the firm's pricing, P.

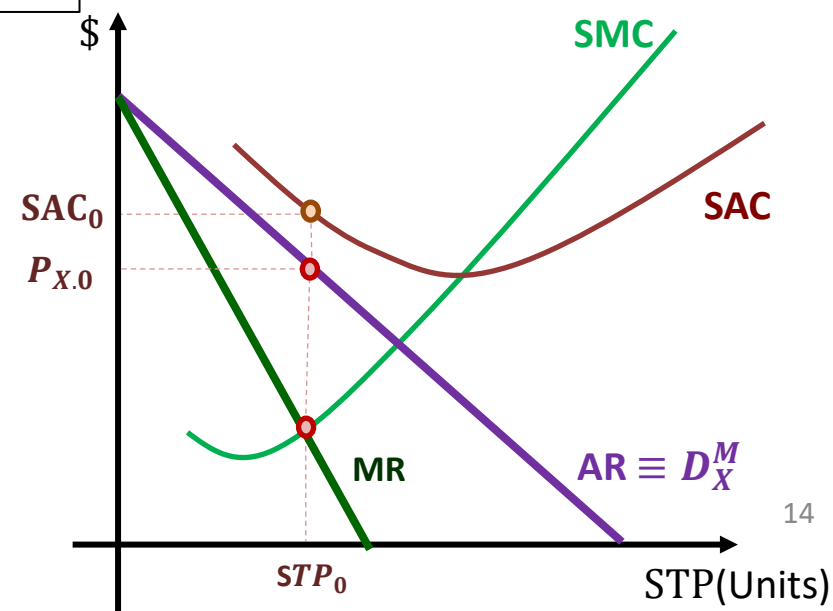
- The SMC curve corresponds to an SAC curve that passes through its lowest point; the SAC corresponding to the above STP will be used to determine whether STPf is positive.

- In the right diagram, Case 1 shows 為 $P_{X.0} > \text{SAC}_0$, so $\text{STPf} > 0$; Case 2 shows $P_{X.0} < \text{SAC}_0$, so $\text{STPf} < 0$. Therefore, the firm in Case 1 will continue to operate! But will the firm in Case 2 exit the market immediately?

Case 1

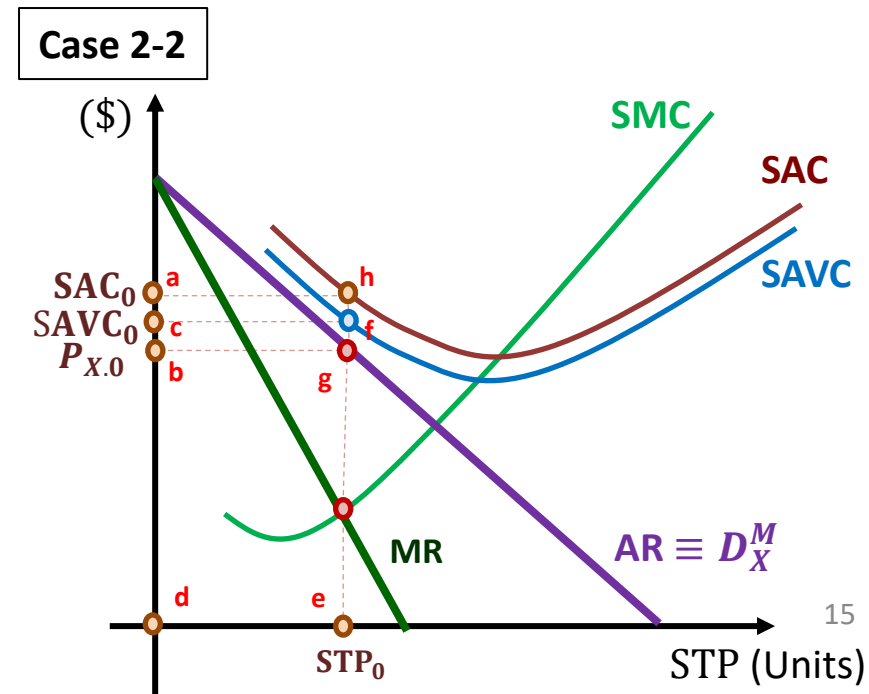
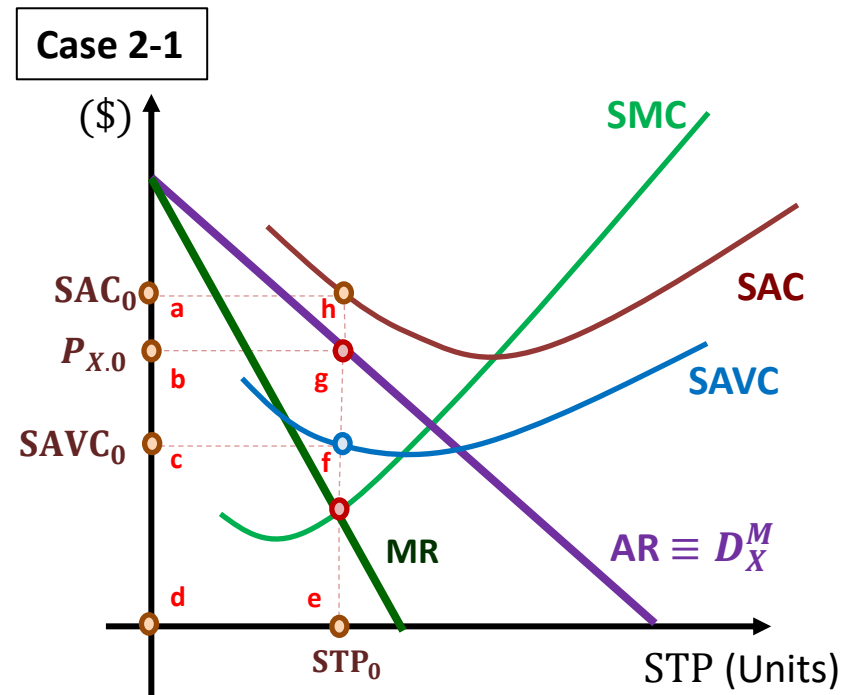


Case 2



$$\begin{aligned} \text{STPf} &= (P - \text{SAC}) \times \text{STP} \\ &= (P - (\text{SAFC} + \text{SAVC})) \times \text{STP} \end{aligned}$$

- In the right diagram, Case 2-1: If the firm produces STP_0 , its total variable cost (STVC_0) is represented by area cdef, where $\text{STVC}_0 = \text{STP}_0 \times \text{SAVC}_0$; the total fixed cost (STFC_0) is represented by area acfh; the short-run total cost (STC_0) is represented by area adeh; and the total revenue (TR_0) is represented by area bdeg.
- In Case 2-1, if the firm produces, the total loss is represented by area abgh. If it does not produce, the total loss is limited to the already incurred STFC_0 , represented by area acfh. Since area abgh < area acfh, the firm should continue to stay in the market and produce.
- In Case 2-2, if the firm produces, the total loss is represented by area abgh. If it does not produce, the total loss is represented by area acfh. Since area abgh > area acfh, the firm should stop production.
- Therefore, even if $\text{STPf} < 0$, when the market price of the product is greater than the average variable cost (Case 2-1), the firm will still continue production in the short run. However, if the market price of the product is less than the average variable cost (Case 2-2), the firm will cease production.



• A monopolist's optimal decision in the short run (LTP and P)

- In the long run, as long as the firm's long-run total profit (LTP_f) > 0 , the firm will continue production; if $LTP_f \leq 0$, the firm will cease production.

$$- LTP_f = (P - LAC) \times LTP$$

Therefore, in the long run, in a monopoly market, if the firm's optimal output, determined by $MR = LMC$, corresponds to LTP and its associated optimal price $P \leq LAC$, the firm will exit the market.

Perfectly Competitive Market

7. In the short run, as long as the market price PPP of the product is still greater than the SAVC of producing the product, the firm will continue to stay in the product market and produce. Although $P > SAVC$ does not guarantee that $STP_f > 0$, it ensures that continuing production is better than stopping production in the short run!

8. In the long run, all firms in the market can adjust their input of production factors to achieve profitability. However, since there are no barriers to market entry, all firms making such adjustments will attract new firms to enter the market. The resulting increase in supply will eventually cause $P = LAC$, making $LTP_f = 0$.

Monopoly

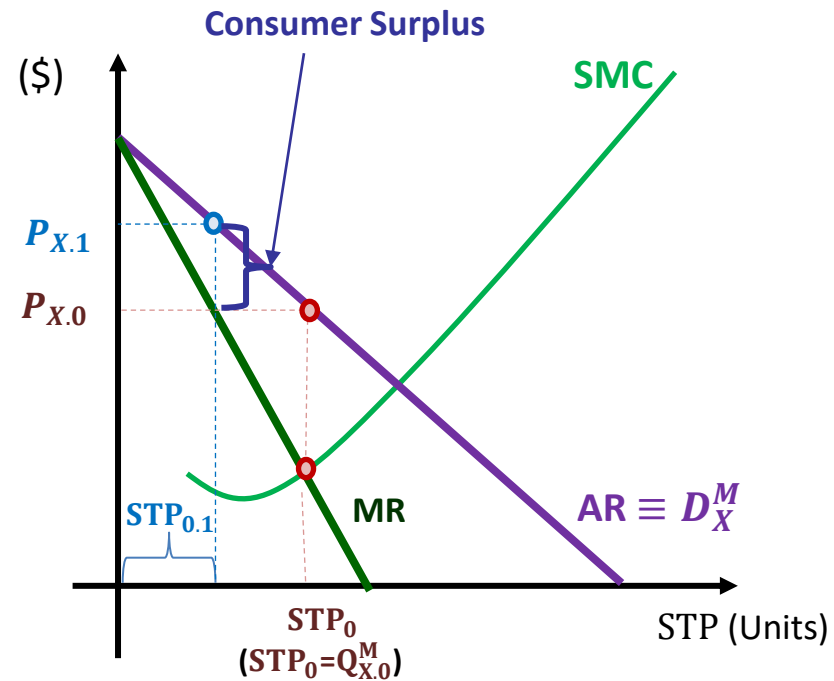
8. In the long run, if the optimal pricing (P) of a monopolist in the market is $P \leq LAC$, the firm will cease production and exit the market. Otherwise, it will remain in the market and continue production.

7-3: Price discrimination by a monopolist

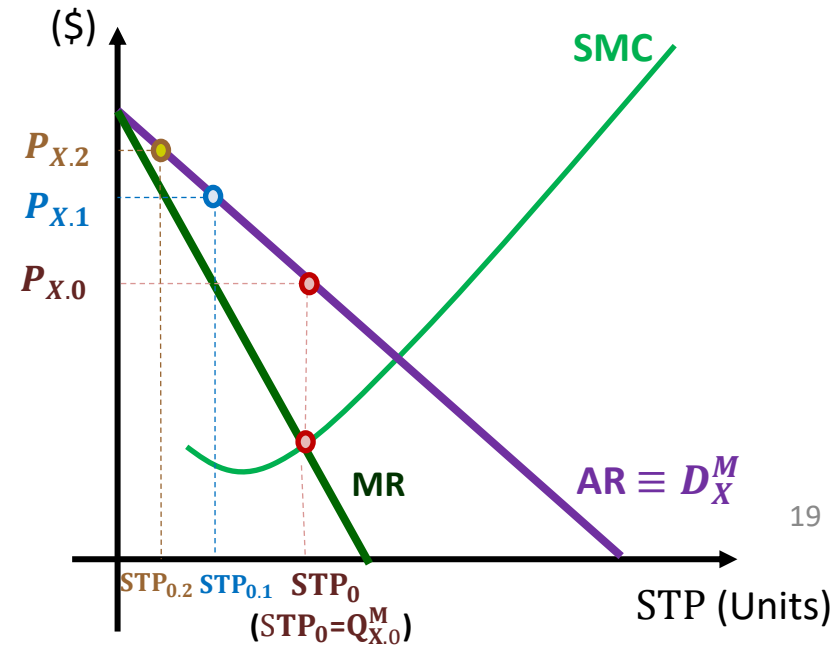
- The firm's objective is: Profit maximization.
- For a monopolist with pricing power, a single pricing strategy is not yet the optimal strategy.

- The previous section discusses monopolists using a single pricing strategy, where the monopolist sets the same product price regardless of the characteristics of market demanders, such as wealth, preference for the product, geographical factors, or the quantity of product purchased.
- In reality, monopolists often use price discrimination to increase their profits. For example, pricing tomatoes at 20 dollars per kilogram or 50 dollars for three kilograms is a type of price discrimination based on purchase quantity.
- The type of price discrimination we are discussing refers to a monopolist setting different unit prices for different customers or different purchase quantities under the same cost structure.

- The premise for price discrimination to exist is that market consumers experience consumer surplus during procurement.
- In the right diagram, $P_{X,0}$ is the optimal price set by the monopolist, based on the market demand curve for product X. Within the market's total demand $Q_{X,0}^M$, if a consumer wishes to purchase a certain portion, for example $STP_{0,1}$, the price they are willing to pay, according to the shape of the demand curve $D_{X,0}^M$, is actually higher than $P_{X,0}$, such as $P_{X,1}$.

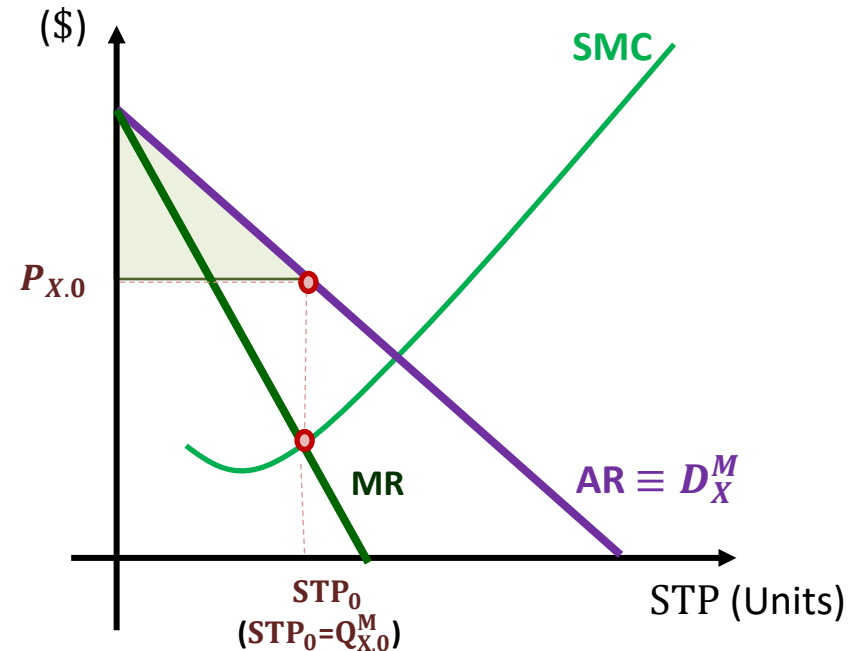


- **Consumer Surplus:** The difference between the price the consumer is willing to pay and the price they actually pay.
- Given the market's total supply $Q_{X,0}^M = STP_0$, if a consumer only buys $STP_{0,2}$ from it, their consumer surplus will be higher than if they buy $STP_{0,1}$. This is because the marginal utility derived from consumption decreases as the quantity purchased increases, resulting in a lower willingness to pay.



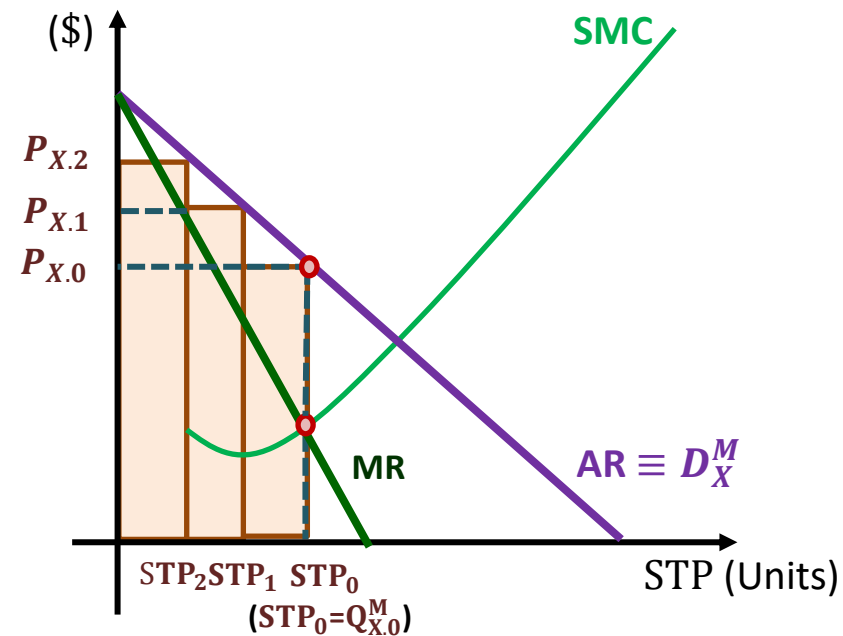
- First-Degree Price Discrimination :

- Given that the firm's optimal output is STP_0 , and $STP_0 = Q_{X,0}^M$, under the condition that the monopolist has pricing power, it can use **perfect price discrimination** by setting prices based on each consumer's demand, to capture the entire market's consumer surplus (the light green area).
- For a monopolist, perfect price discrimination maximizes profits. If it can be done, this is the most ideal situation. However, in reality, it is rarely achieved.
- We call perfect price discrimination **first-degree price discrimination**.

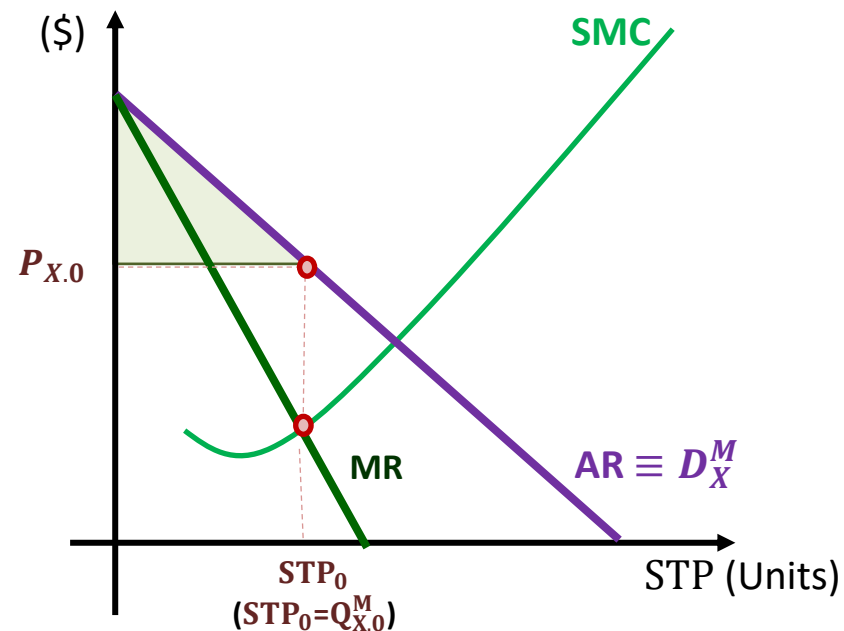


- Second-Degree Price Discrimination :

- Also known as block pricing, the firm sets product prices based on different purchase quantity intervals for consumers.
- For example, in the diagram above, if the consumer's purchase quantity is within STP_2 , the firm sets the price at $P_{X.2}$. If the consumer's purchase quantity reaches STP_1 , the firm charges $P_{X.1}$ for the additional portion.
- Block pricing deprives part of the consumer surplus.

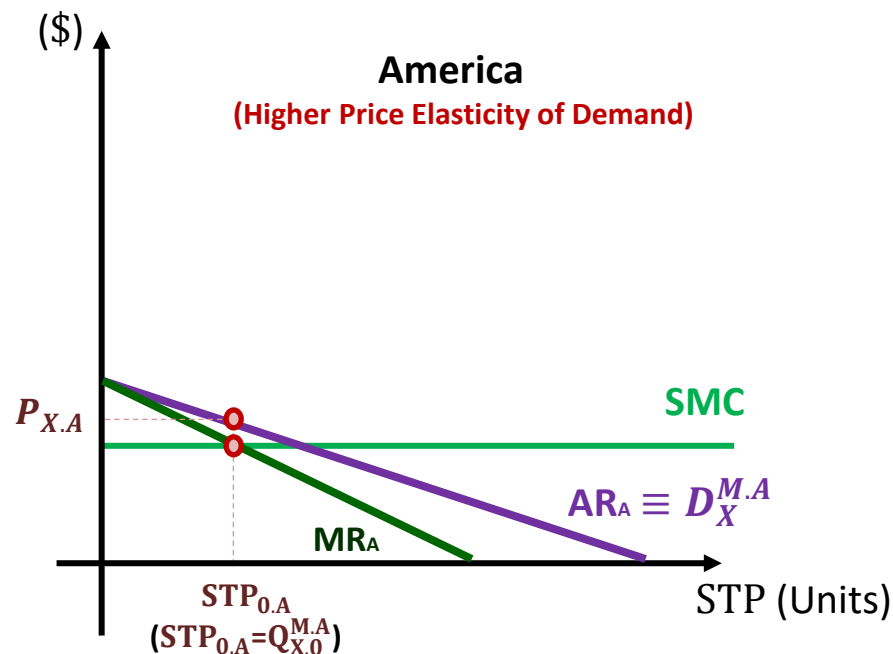
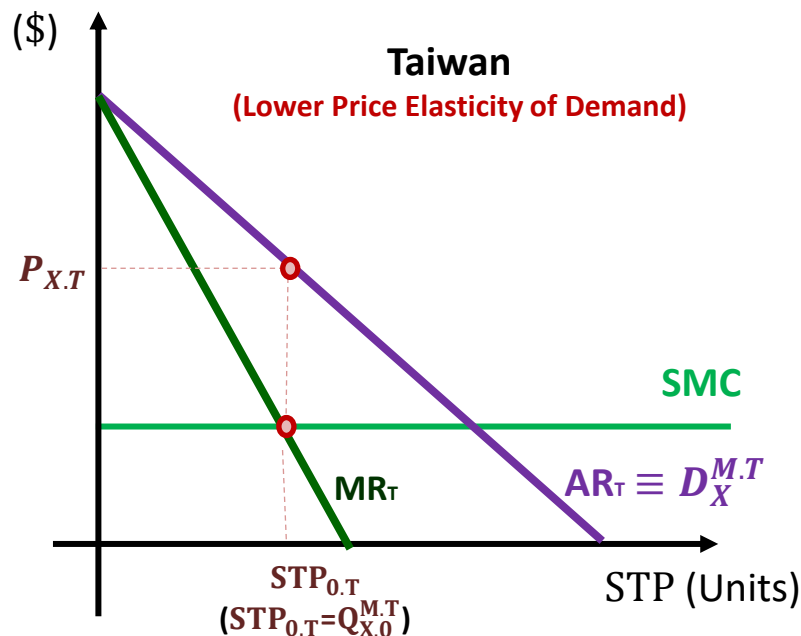


Consumer surplus is partially deprived



Consumer surplus is totally deprived

- Third-Degree Price Discrimination :



- Assume a monopolist has a marginal cost (SMC) that is a horizontal line, and the SMC is the same for the firm in both Taiwan and the United States. Because the demand curves in the two locations are different, they correspond to different MR lines. Under the optimal output decision where $MR = SMC$, the prices set by the firm in the two locations will also differ.
- If the firm's SMC is the same in both locations, and $SMC = MR_T$ and $SMC = MR_A$, then to maximize its total profit (STPf) in both locations, the pricing strategy will follow $SMC = MR_T = MR_A$, corresponding to their respective prices.
- A monopolist employing third-degree price discrimination will set lower prices in markets with higher price elasticity of demand and higher prices in markets with lower price elasticity of demand.

- **Recall:**

When the price elasticity of demand is smaller, if the product's price decreases, the total revenue of the firm will decrease. Additionally, regardless of whether the price is high or low, the change in demand will not be significant. If the product lacks substitutes, the firm is more likely to benefit by increasing the selling price.

Therefore:

The implication of this third-degree price discrimination is that a profit-maximizing monopolist tends to adopt a high-pricing strategy in markets where the price elasticity of demand is initially low.

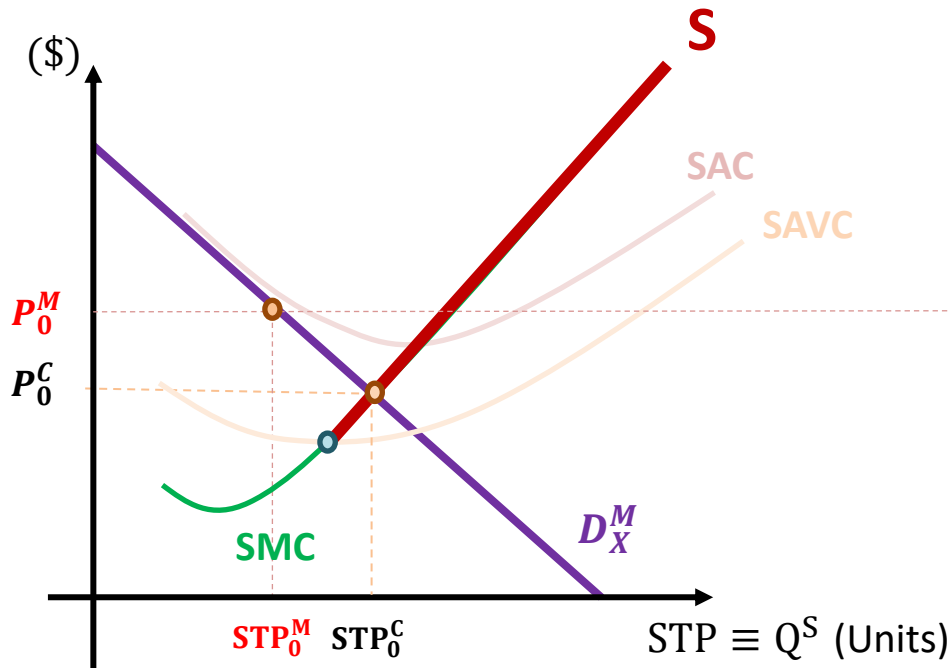
- Another type of third-degree price discrimination is spatial price discrimination. When a monopolist produces a product with identical demand (the same demand curve) in two locations but faces different transportation costs to deliver the product to those locations, the firm will set different prices for the product in the two locations accordingly.

- Price discrimination must meet the following three prerequisites:
 - (a) The existence of consumer surplus
 - (b) The discussion pertains to the same type of product
 - (c) The monopolist's cost structure (i.e., STC, SAC, and SMC) is identical
- To make price discrimination effective, the following conditions must be met:
 - (a) Consumers must be able to be segmented into groups
 - If consumers can collude or move freely between markets (e.g., the U.S. and Taiwan), the firm's strategy for consumer segmentation may fail. Once segmentation fails, price discrimination cannot be implemented.
 - (b) The product must not be resalable
 - If a product can be purchased in market A and sold in market B, the monopolist's price discrimination strategy will also fail, as consumers will only buy from the lower-priced market and avoid consuming in the higher-priced market.
 - Generally speaking, services are less likely to be resold, making it easier to implement price discrimination.

7-4: Two additional comparisons between monopolists and perfectly competitive firms

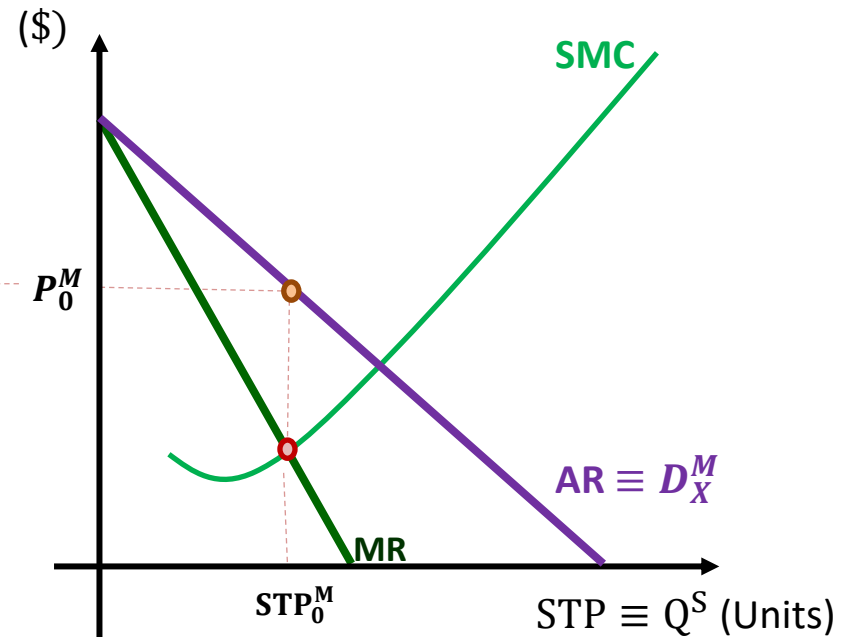
- For ease of comparison, we assume that the two firms have identical conditions (e.g., the same cost structure), but one is a monopolist, and the other operates in a perfectly competitive market
- Both firms face the same market demand curve

Perfectly Competitive Market



9. The supply curve of a certain firm is S . When the firm faces the market demand curve D_X^M and the market price of the product is P_0^C the firm will determine the optimal output using the principle $P_0^C = MR = SMC$, resulting in an output of STP_0^C .

Monopoly



9. The firm does not have a supply curve, but it has pricing power. When it faces the market demand curve D_X^M , the firm will determine the optimal output STP_0^M based on the principle $MR = SMC$, and then set the price P_0^M on D_X^M corresponding to that output.

When the two firms face same demand curve,
 $STP_0^C > STP_0^M$; $P_0^C < P_0^M$

Perfectly Competitive Market

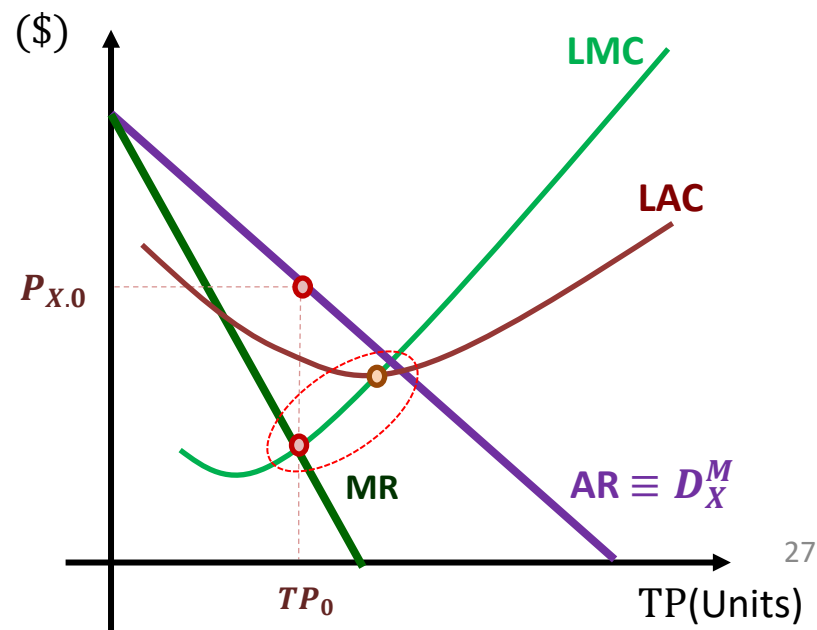
Monopoly

Recall: The LAC and LMC curves are composed of numerous SAC and SMC curves

10. In the long run, firms in a perfectly competitive market will choose to produce at the lowest LAC corresponding to TP under the principle $P = MR = LMC = LAC$. This is the most production-efficient method of production, but it also results in $LTP_f = 0$.

When facing the same market demand curve, firms in a perfectly competitive market are more production-efficient (they will produce at the TP where LAC is at its lowest)!

10. In the long run, if a monopolist produces and prices based on the principle $MR = LMC$, the intersection of the MR and LMC lines is unlikely to precisely align with the lowest point of the LAC curve.



7-5: Policies to Restrain Monopoly

Advantages of a Monopolist:

- Monopolists achieve economies of scale (LAC decreases as TP increases), known as natural monopoly.
- Governments can promote research and development by granting patent protection to establish monopolists.

Disadvantages of a Monopoly Firm:

- Monopolists can use perfect price discrimination to take away all consumer surplus
- Monopolists reduce economic surplus
- Monopoly firms are not production-efficient (they do not produce at the lowest LAC)

- **Anti-Monopoly Law (Antitrust Law):**
 - Policy Objectives of Antitrust Law:
 - (1) Restrict the abuse of monopoly power
 - (2) Prevent unfair competition practices
 - In Taiwan, the Fair Trade Act serves as the relevant law. The central competent authority is the Fair Trade Commission of the Executive Yuan. The Fair Trade Act regulates the following: (1) Collusion (2) Unfair competition

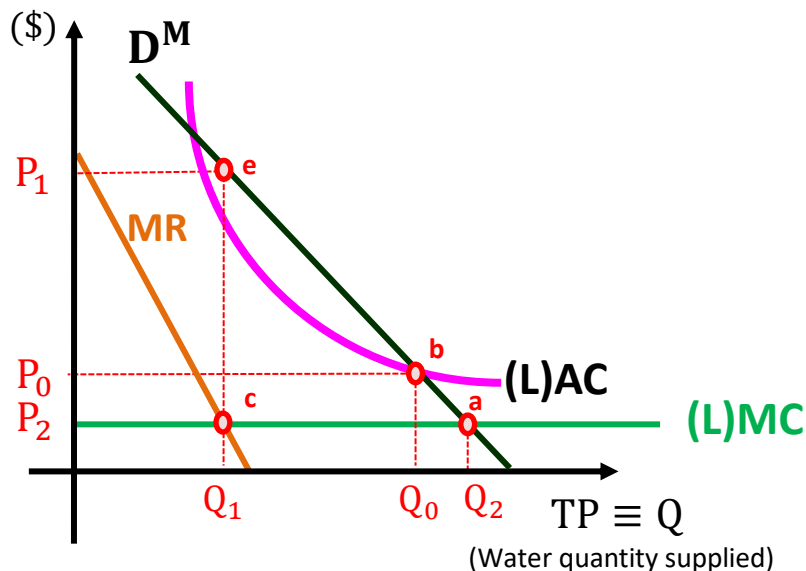
- **Impose entry barriers:**

- Some industries experience decreasing average costs as total production increases, forming natural monopolies, such as electricity, water supply, postal services, cable television, etc.
- Governments often impose market entry barriers for natural monopoly industries:
 - **Only government monopoly operation is permitted, such as water supply and postal services.**
 - **Private operation is allowed, but the number of private firms is restricted, such as public transportation and cable television.**
- The reason governments set market entry barriers for natural monopoly industries is to avoid excessive production costs caused by multiple firms entering the market simultaneously. Thus, regulatory measures align with economic efficiency.
 - **It may also be due to an excessive number of firms, leading to inefficiencies in the product market where the total production becomes too high, preventing economies of scale.**
- Regardless of whether the government chooses public or private monopoly markets, it cannot guarantee that these monopolist will operate more efficiently than those excluded from the market. If they do, entry barriers should be established.

- **Price regulation:**

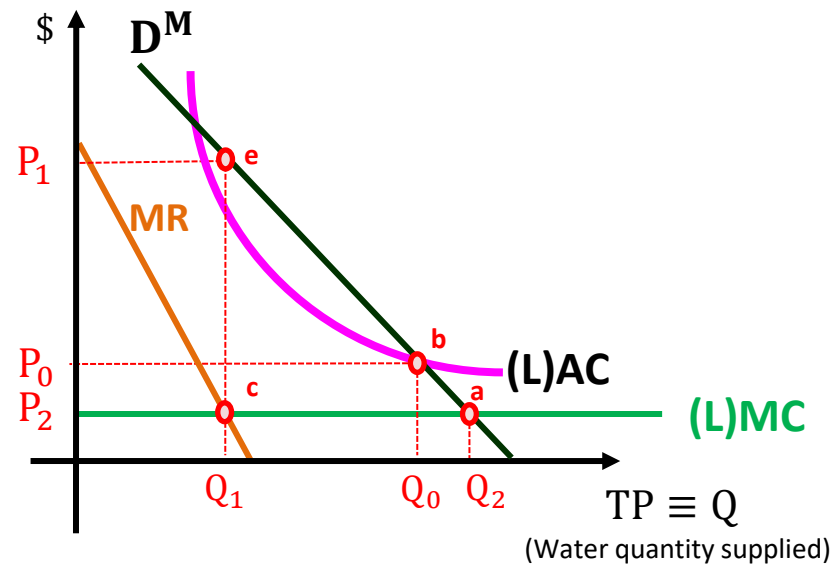
- However, in many cases, the natural monopolies created by government-imposed entry barriers are actually a result of flawed policies, as they do not lead to higher operational efficiency
- Moreover, natural monopolists, due to their monopolistic position, seek excessive returns. As a result, their profit-maximizing production decisions often lead to a total output that is actually lower than optimal.

- Assume the graph on the left represents the supply and price of a water company.



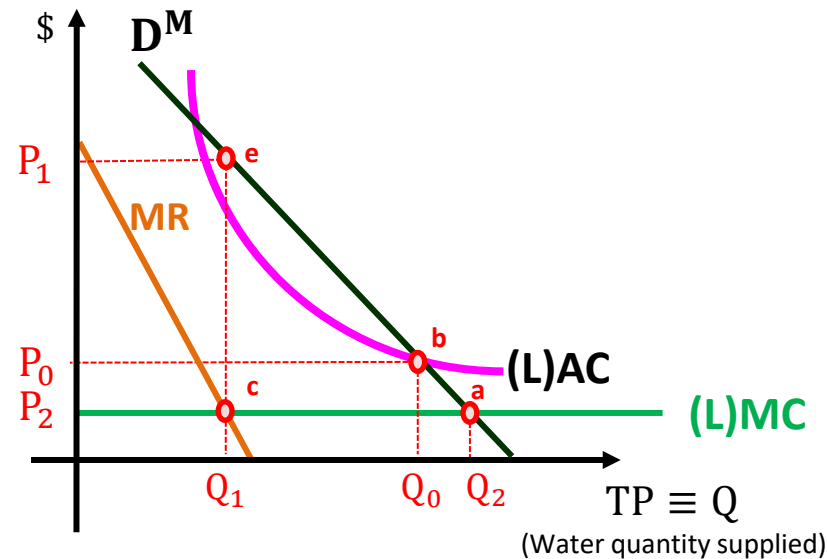
- Without price regulation, the profit-maximizing water company will choose the water quantity supplied Q_1 where $(L)MC = MR$, and then determine the price P_1 based on D^M .
- $P_1 > MC$ indicates that the highest price consumers are willing to pay exceeds the marginal cost of the water company's last unit of production. This price difference also suggests that, in reality, consumers demand more water. However, under the firm's profit-maximizing decision, it only produces up to Q_1 , leading the firm to earn profits ($P_1 > (L)AC$), but the market never reaches supply-demand equilibrium. As a result, consumers must pay a higher price to access water (market failure).

- **Price regulation – Pricing using marginal cost**



- To prevent the market failure described above, the government can use marginal cost pricing to regulate firms.
- The government can directly require the water company to increase production to the point where (long-term) marginal cost equals the quantity corresponding to the highest price consumers are willing to pay. In the graph above, this corresponds to point **a** and the output Q_2 . At this point, the price would be reduced to P_2 .
- However, since $P_2 < (L)AC$, the firm would incur losses. In the long run, the water company may find this unsustainable. The government could subsidize the water company, but if the subsidy is funded by taxing consumers, it would inevitably affect other economic activities of consumers, which may not truly align with economic efficiency.

- **Price regulation – Pricing using average cost**



- Compared to the pricing using marginal cost, the government might consider a pricing method where the monopolist just breaks even – the average cost pricing method. This approach sets the (long-run) average cost equal to the quantity corresponding to the highest price consumers are willing to pay. In the graph above, this corresponds to point **b** and the output Q_0 .
- Although Q_0 is slightly less than Q_2 , it is still significantly higher than the output Q_1 , thus helping to prevent market failure!